Addition and subtraction of algebraic terms

1

a $4x + 4y$

b $15m + 19n$

c $3x + 2y$

d $25m + 8n$

e $21xy$

f $8q$

g $20p^2 - 11p$

h $3n^2 - 4m + 2m^2$

i $3x + 19y$

j $18a - 8b$

k $7x^2 + x$

l $8x - 7y + 4z$

m $25y - 9y^2$

n $11a - 7b + 4c$

o $3ab + 2c$

p $23j + 2k$

q $25x + 2y$

r $12y$

s $5m + 12n$

t $6s - 2t$

2

a $-3x - y + 10z$

b $-6xy + 2yz + 6xz$

c $12x^2 + 9x + 5$

d $-16x + 10y + 41$

e $11mn - 2m$

f $24x^2 + 5x - 15$

g $st + 7s + 7t$

h $19ab + 3bc - 9b$

i $8x + 6y$

j $7p + 7$

k $2pq + 11p^2q$

l $17x^2$

m $20xyz + 23xy$

n $-12x - 4y$

o $-6rk + 3r - 6k$

p $-14vw - 2v^2w$

q $-5n + 7$

r $-2n + 5$

Multiplication of algebraic terms

3

a $60u$

b $-21u$

c $10x$

d $33y$

e $-72x$

f $24u$

g $-6w$

h $-30y$

4

a x^2

b $6x^2$

c $4y^2$

d $15a^2$

e $16x^2y$

f $27b^2$

g $100a^2$

h $16x^2$

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5

a $72mn$ b $24wz$ c $50rs$ d $54rs$

e $42u^2v^8$ f $224p^3q^3$ g $8ab$ h $-21wz$

i $20rs$ j $-60r^8s^7$ k $-12p^5q^5$ l $-10a^4b^3$

6

a $96rs$ b $630wxy$ c $-80wxy$ d $-480hkrs$

e $-90xy$ f $-80a^3$ g $90x^2y$ h $-40hijk$

Division of algebraic terms

7

a x b $3v$ c $\frac{m}{4}$ d $\frac{1}{4}$

e xy f $7q$ g $\frac{4n}{5}$ h $\frac{6}{w}$

i $\frac{1}{q}$ j $10v^4$ k $-6a$ l -11

m $-\frac{1}{11}$ n -4 o $\frac{4}{9}$ p $-ac$

q $-\frac{1}{j}$ r $-\frac{6}{k}$ s $-\frac{1}{b}$ t $-\frac{2}{3}$

u $-3w^5$ v 12

8

a $\frac{m}{8}$ b 5 c 2 d -4

e $-11s$ f 6 g $2n$ h $3x$

i $3r$ j -2 k $11x$ l $10c$

9

a 6 b 2 c $3x^2$ d 14

e $3a$ f $\frac{9}{2}$ g $18x$ h $-4x$

i $33xy$ j $15y^2$ k $10x$ l y

Applications

10 $(10 + a)$ cm

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11

a 24 **b** $6y$

12

a $3x$ cm

b The water level is increasing by the same amount each hour.

13 $5p + 10q$

14

a $(300 + 70t)$ m

b Uther is paddling at a constant pace over the first t minutes.

15

a $9 + y$

b It represents that it takes Sally 9 minutes longer than the bus trip to get to school.

16

a $44WL$ cm^3 **b** $132WLH$ cm^3 **c** $3H$ **d** 264

17

a $6xy$ cm^3 **b** $48xyz$ cm^3 **c** $8z$ **d** 120

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